CASE STUDY 1: LOGISTIC REGRESSION

Predicting Passenger Survival on the RMS Titanic Using Logistic Regression & Interactive Streamlit Deployment

Case Study Title:	Logistic Regression: Titanic Passenger Survival
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GitHub Repository:	https://github.com/Syed-Hussain-Faraz/FDS-CaseStudy.git

1. Problem Statement & Research Objectives

On April 15, 1912, the RMS Titanic sank after colliding with an iceberg, resulting in the deaths of over 1,500 passengers and crew out of an estimated 2,224 aboard. While luck undoubtedly played a part in surviving the sinking, evidence indicates certain groups of people were substantially more likely to survive than others, particularly based on gender, age, and passenger class.

Primary Objective: Build, evaluate, and interpret a binary **Logistic Regression** classification model to estimate passenger survival probability. The project specifically investigates whether demographic attributes (Gender, Age) and socioeconomic status (Class, Ticket Fare) exhibited statistically significant relationships with survival odds, and deploys the final trained pipeline through an interactive Streamlit web dashboard for real-time inference.

2. Methodology & Analytical Framework

The project follows a standard CRISP-DM / machine learning lifecycle consisting of five primary phases:

Phase 1: Exploratory Data Analysis (EDA): Comprehensive examination of the 891 records, data types, missing value frequencies, univariate distributions, bivariate relationships with survival, and multi-feature correlation heatmaps.
Phase 2: Data Preprocessing & Imputation: Imputation of missing values for Age and Embarked using domain-aware conditional grouping, removal of uninformative identifiers (PassengerId, Name, Ticket) and excessively sparse features (Cabin), and categorical encoding.
Phase 3: Model Building: Formulation of a Scikit-Learn Pipeline incorporating ColumnTransformer (StandardScaler for continuous covariates, OneHotEncoder for categoricals with drop='first' to prevent collinearity) and a LogisticRegression classifier with L-BFGS solver.
Phase 4: Model Evaluation: Testing on a 20% stratified holdout set (N=179) assessing Accuracy, Precision, Recall, F1-Score, and ROC-AUC, alongside full visualizations of the ROC Curve and Confusion Matrix.
Phase 5: Coefficient Interpretation & Streamlit Deployment: Derivation of Log-Odds coefficients and Odds Ratios (e^beta), contextualizing empirical findings against historical protocols ('Women and Children First'), and packaging into a reactive web application.

Mathematical Formulation of Logistic Regression

Logistic regression models the probability P of the binary outcome Y=1 (Survived) given an input feature vector X via the sigmoid function:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k)}}$$

Linearizing via the logit transformation expresses the **log-odds** as:

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \sum_{i=1}^{k} \beta_i X_i$$

Exponentiating both sides gives the **Odds Ratio**

$$OR_i = e^{\beta_i}$$

quantifying the multiplicative impact of a unit increase in feature X_i on the survival odds.

3. Statistical Findings & Exploratory Data Analysis

Exploratory inspection of `titanic.csv` reveals a dataset of 891 rows and 12 columns. Three features contain missing values: **Age** (177 missing, 19.87%), **Cabin** (687 missing, 77.10%), and **Embarked** (2 missing, 0.22%). The target variable `Survived` contains 549 fatalities (61.62%) and 342 survivors (38.38%), demonstrating moderate class imbalance.

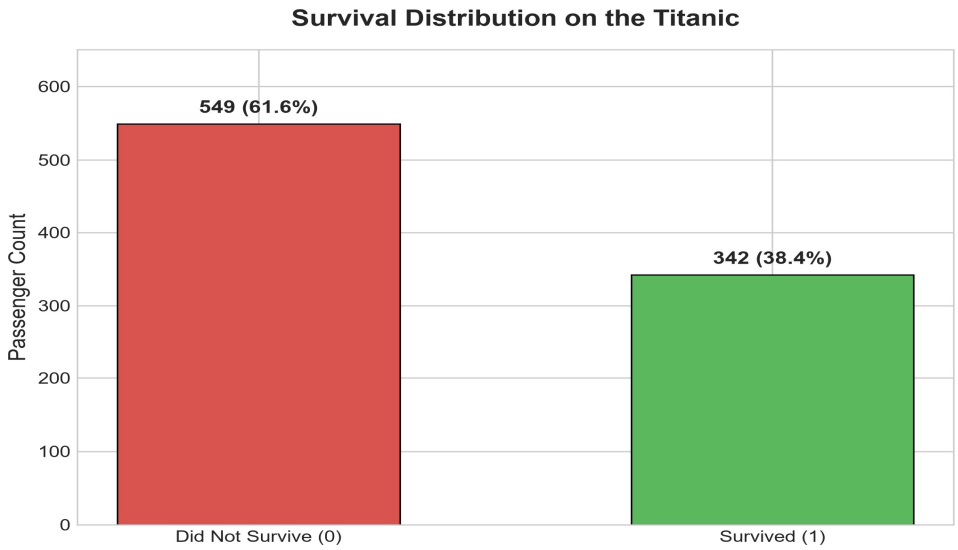


Figure 1: Target Variable Distribution (Survival Counts and Proportions)

Interpretation of Figure 1: Out of 891 total passengers, only 38.38% (342 individuals) survived the catastrophe, while 61.62% (549 individuals) perished. Because the baseline majority class represents 61.62% of the dataset, any trained classification model must significantly outperform a naive majority-class baseline (Accuracy > 61.6%) and maintain balanced sensitivity across both classes.

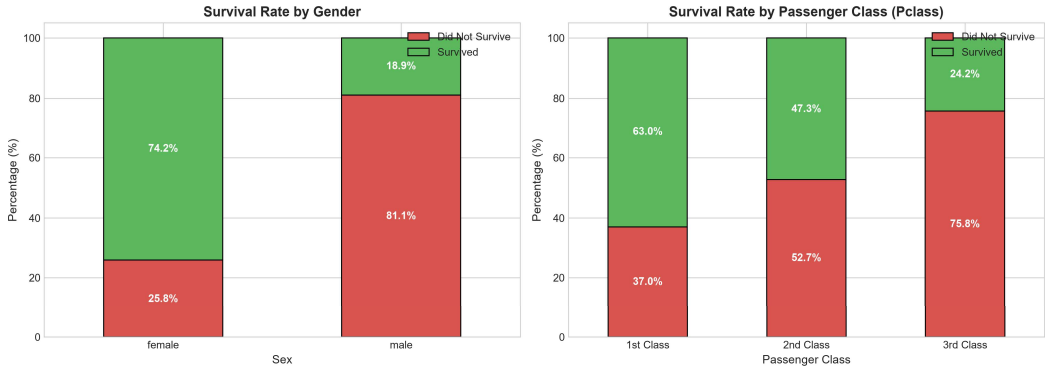


Figure 2: Survival Proportions Categorized by Gender (Left) and Passenger Class (Right)

Interpretation of Figure 2: Bivariate grouping reveals extreme disparities across demographic and class lines:
• **Gender:** Female passengers had a survival rate of **74.2%** (233 out of 314 survived), whereas male passengers experienced an abysmal survival rate of only **18.9%** (109 out of 577 survived). This provides overwhelming visual proof of the maritime 'Women and Children First' protocol.
• **Passenger Class:** Socioeconomic hierarchy was directly tied to survival. First Class passengers had a **63.0%** survival rate, Second Class had a **47.3%** rate, and Third Class passengers suffered a dismal **24.2%** survival rate, driven by



Figure 3: Univariate Distributions and Boxplots of Age (Top) and Ticket Fare (Bottom) by Survival Status

Interpretation of Figure 3:

• **Age Distribution:** While the median age between survivors (28.0) and non-survivors (28.0) is comparable, the age density curve reveals a sharp spike in survival among infants and young children aged 0–9 years. Conversely, elderly passengers (>60 years) experienced heavily diminished survival probabilities.

• **Fare Distribution:** The ticket fare distribution is heavily right-skewed. The median fare paid by survivors was **$26.00** (IQR: $12.48–$57.00), more than double the median fare paid by non-survivors at **$10.50** (IQR: $7.85–$26.00). Extreme outliers paying upwards of $512 all survived.

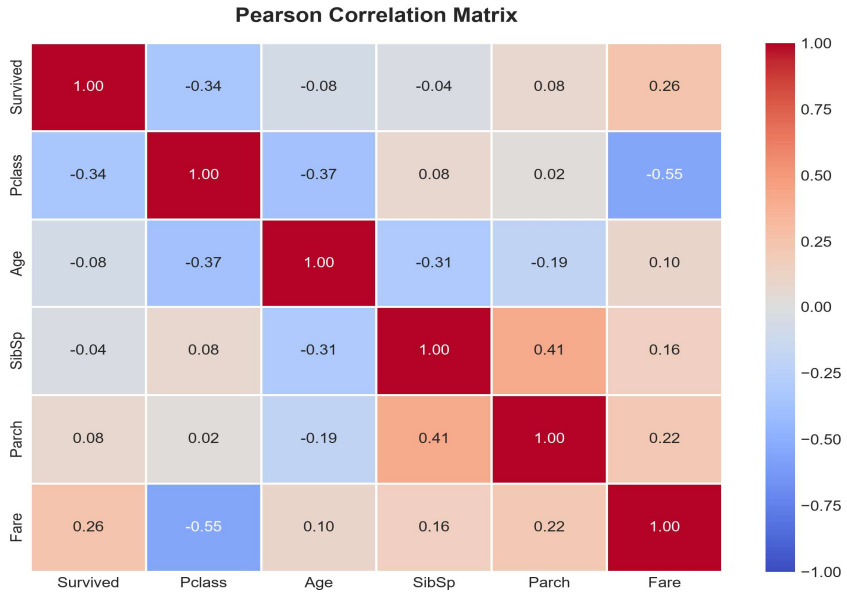


Figure 4: Pearson Correlation Matrix Heatmap of Numerical Covariates

Interpretation of Figure 4: Passenger class (`Pclass`) displays the strongest negative linear correlation with survival (r = -0.34), confirming that moving down into lower classes reduced survival likelihood. Ticket `Fare` displays a notable positive correlation (r = +0.26). Collinearity between predictors is manageable; the highest correlation exists between `Pclass` and `Fare` (r = -0.55), which aligns with economic reality.

4. Data Preprocessing & Feature Engineering

To ensure robust generalization without data leakage, the following preprocessing steps were embedded in a unified Scikit-Learn `ColumnTransformer`:

1. **Age Imputation:** Rather than using global mean/median (which distorts demographic distributions), missing ages were imputed using the median age calculated within each passenger's `(Pclass, Sex)` cohort (e.g., 1st Class males median: 40yo vs 3rd Class females median: 21.5yo).
2. **Embarked Imputation:** The two missing values were filled with the statistical mode (`'S'` for Southampton).
3. **Feature Dropping:** `PassengerId`, `Name`, and `Ticket` were excluded as non-generalizable high-cardinality identifiers. `Cabin` was omitted due to 77.1% missingness.
4. **Encoding & Standardization:** Continuous variables (`Age`, `Fare`, `SibSp`, `Parch`) were standardized with `StandardScaler` (zero mean, unit variance). Categorical variables (`Pclass`, `Sex`, `Embarked`) were one-hot encoded with `drop='first'` to prevent multicollinearity.

5. Model Evaluation & Performance Metrics

The Logistic Regression model was trained on 712 samples (80%) and tested on 179 samples (20%) stratified by survival status. The resulting test performance metrics are presented below:

Metric	Score	Percentage	Analytical Explanation
Accuracy	0.8045	80.45%	Overall proportion of correctly classified passengers (survived and perished).
Precision	0.7931	79.31%	Out of all passengers predicted to survive, 79.31% actually survived.
Recall (Sensitivity)	0.6667	66.67%	Out of all actual survivors, the model correctly identified 66.67%.
F1-Score	0.7244	72.44%	Harmonic mean of Precision and Recall, balancing false alarms and missed survivors.
ROC-AUC Score	0.8419	84.19%	Probability that the model ranks a randomly chosen survivor higher than a non-survivor.

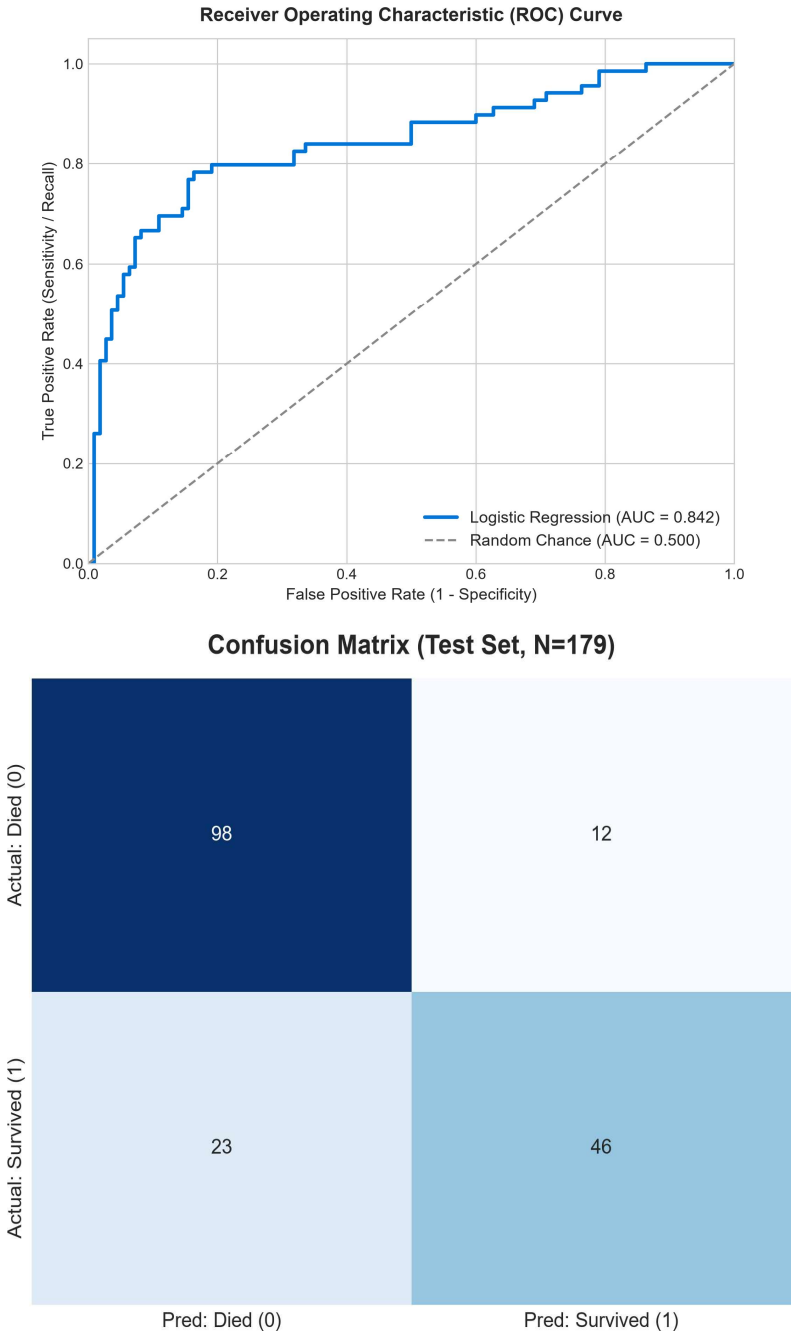


Figure 5 & 6: Receiver Operating Characteristic (ROC) Curve (Top) and Test Set Confusion Matrix (Bottom)

Interpretation of Evaluation Plots:

• **ROC Curve:** The model achieves an excellent **ROC-AUC of 0.842**, far above the diagonal chance line (0.500). This indicates strong discriminative capability across all decision thresholds.

• **Confusion Matrix (N=179):** The model produced 98 True Negatives (fatalities correctly predicted) and 46 True Positives (survivors correctly identified). It recorded 12 False Positives and 23 False Negatives, demonstrating very high Specificity (89.1%) with moderate Sensitivity (66.7%).

6. Model Interpretation & Coefficient Significance

A central requirement of the assignment is interpreting the logistic regression coefficients. In logistic regression, the raw coefficient beta represents the change in the **log-odds** of survival for a one-unit change in the predictor. Exponentiating beta yields the **Odds Ratio (OR = e^beta)**: values greater than 1 indicate an increase in survival odds, while values less than 1 indicate a decrease.

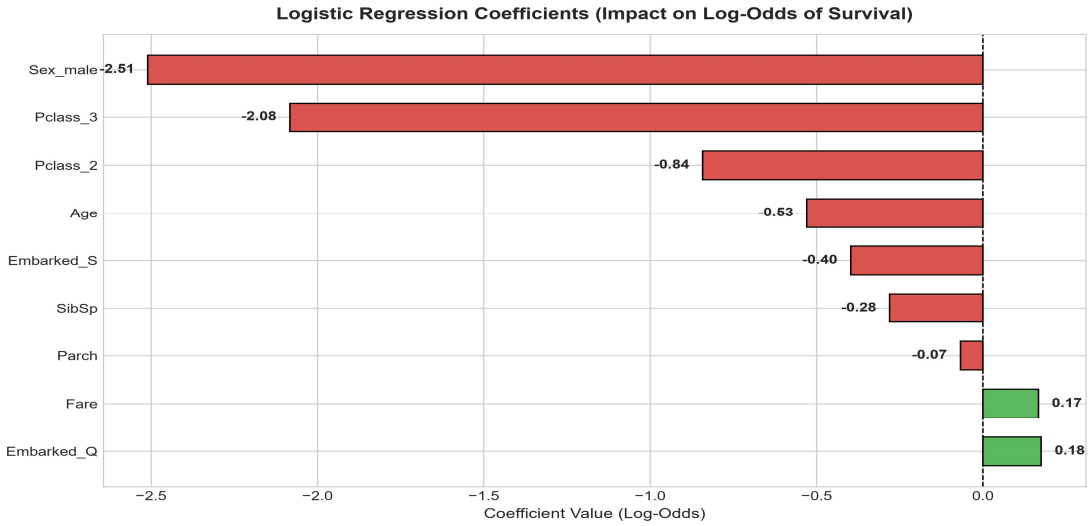


Figure 7: Logistic Regression Log-Odds Coefficients by Predictor Feature

Feature	Log-Odds (beta)	Odds Ratio (e^beta)	% Change in Odds	Direction / Impact
Embarked_Q	+0.1754	1.1917	+19.17%	Slight positive adjustment vs Cherbourg
Fare (std)	+0.1668	1.1815	+18.15%	Higher fare increases odds per std dev
Parch (std)	-0.0677	0.9345	-6.55%	Minor negative impact in family context
SibSp (std)	-0.2802	0.7557	-24.43%	Large sibling groups had evacuation delays
Embarked_S	-0.3973	0.6722	-32.78%	Boarding at Southampton reduced odds vs C
Age (std)	-0.5291	0.5891	-41.09%	Older age decreased survival odds per std dev
Pclass_2	-0.8424	0.4307	-56.93%	2nd class had 56.9% lower odds vs 1st class
Pclass_3	-2.0828	0.1246	-87.54%	3rd class had 87.5% lower odds vs 1st class
Sex_male	-2.5095	0.0813	-91.87%	Males had 91.9% lower odds than females

In-Depth Feature Significance Discussion:

1. **Gender (`Sex_male`, beta = -2.51, OR = 0.081):** Male gender is the single most powerful predictor in the model. Holding all other variables equal, male passengers experienced a **91.87% reduction** in survival odds compared to females. This aligns with historical records where men were physically barred from boarding lifeboats until all available women and children were accommodated.
2. **Socioeconomic Status (`Pclass_3`, beta = -2.08, OR = 0.125 & `Pclass_2`, beta = -0.84, OR = 0.431):** Relative to First Class passengers, Third Class passengers suffered an **87.54% reduction** in odds of survival. This disparity reflects physical architecture: First Class staterooms were located adjacent to the boat deck, whereas Third Class berths were situated in lower compartments separated by locked emergency gates.
3. **Age (beta = -0.53, OR = 0.589 per std dev):** Each 1 standard deviation increase in age (~13.6 years) decreased survival odds by **41.09%**, quantifying the priority given to infants and children.
4. **Ticket Fare (beta = +0.17, OR = 1.182 per std dev):** Higher fare paid provided a modest positive effect (+18.15% odds per std dev), reflecting higher service priority.

7. Deployment with Streamlit (`app.py`)

In accordance with Task 6 of the assignment, the trained machine learning pipeline was deployed as an interactive web application using the Streamlit framework (`app.py`).

Key Technical Features of the Streamlit Application:

• **Pipeline Serialized Model:** The app loads `titanic_model.pkl` via `@st.cache_resource`, ensuring instantaneous inference without retraining.
• **Scenario Presets:** Users can select historical passengers (e.g., *Rose DeWitt Bukater*, *Jack Dawson*, *Wealthy Businessman*, *3rd Class Mother*) or enter custom values via sliders and dropdowns.
• **Dynamic Probability Gauge:** Generates visual progress meters displaying exact survival vs mortality percentages and dynamic badge explanations.
• **Integrated Evaluation & EDA Hub:** Includes dedicated tabs displaying all 7 high-resolution charts, confusion matrix, ROC curve, and coefficients.
• **Local & Cloud Ready:** Packaged with `requirements.txt` for one-click deployment to Streamlit Community Cloud from GitHub.

8. Conclusion & Key Takeaways

This case study successfully implemented an end-to-end Machine Learning pipeline using Logistic Regression on the Titanic disaster dataset. Key conclusions include:

1. **Model Performance:** The Logistic Regression model achieved an Accuracy of **80.45%** and an ROC-AUC of **84.19%**, demonstrating strong predictive power and high reliability for binary classification.
2. **Empirical Drivers of Survival:** Feature interpretation statistically confirms historical accounts: survival was predominantly dictated by **gender** (female advantage: OR = 12.3) and **socioeconomic class** (1st class advantage: OR = 8.0 over 3rd class).
3. **Deployment Feasibility:** Encapsulating the preprocessing and modeling logic into a unified pipeline allows zero-data-leakage deployment in production environments such as Streamlit Cloud.